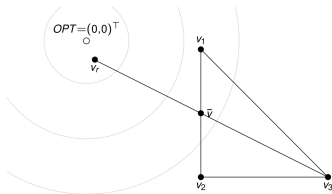
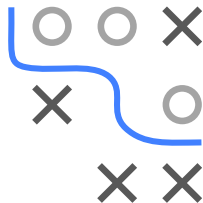


Optimization in Machine Learning

Nelder-Mead method

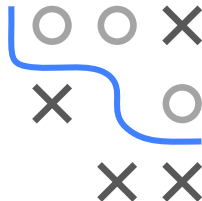


Learning goals

- General idea
- Reflection, expansion, contraction
- Advantages & disadvantages
- Examples

NELDER-MEAD METHOD

- Derivative-free method $\Rightarrow$ heuristic
- Generalization of bisection in d -dimensional space
- Based on d -simplex, defined by $d + 1$ points:
 - $d = 1$ interval
 - $d = 2$ triangle
 - $d = 3$ tetrahedron
 - ...



NELDER-MEAD METHOD

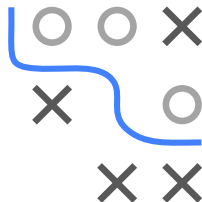
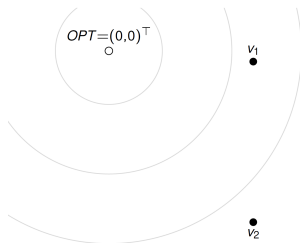
A version of the Nelder-Mead method:

Initialization: choose $d + 1$ random, affinely independent points $\mathbf{v}_i$
($\mathbf{v}_i$ are vertices: corner points of the simplex/polytope)

1. Order points according to ascending function values

$$f(\mathbf{v}_1) \leq f(\mathbf{v}_2) \leq \dots \leq f(\mathbf{v}_d) \leq f(\mathbf{v}_{d+1})$$

with $\mathbf{v}_1$ best point, $\mathbf{v}_{d+1}$ worst point

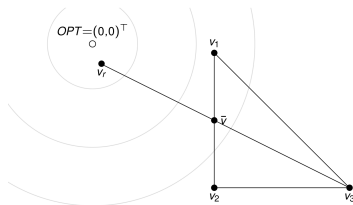


NELDER-MEAD METHOD

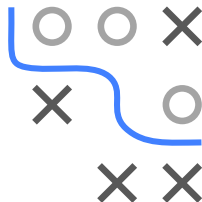
3. Reflection: compute reflection point

$$\mathbf{v}_r = \bar{\mathbf{v}} + \rho(\bar{\mathbf{v}} - \mathbf{v}_{d+1}),$$

with $\rho > 0$. Compute $f(\mathbf{v}_r)$.



Note: Default value for reflection coefficient: $\rho = 1$



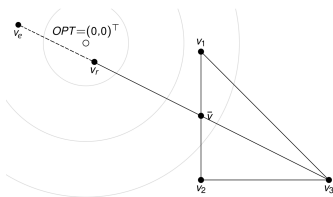
NELDER-MEAD METHOD

Distinguish three cases:

- Case 2: $f(\mathbf{v}_r) < f(\mathbf{v}_1) \Rightarrow$
Expansion:

$$\mathbf{v}_e = \bar{\mathbf{v}} + \chi(\mathbf{v}_r - \bar{\mathbf{v}}), \quad \chi > 1$$

We discard $\mathbf{v}_{d+1}$ and accept the better of $\mathbf{v}_r$ and $\mathbf{v}_e$



Note: default value for expansion coefficient: $\chi = 2$

NELDER-MEAD METHOD

- Case 3: $f(\mathbf{v}_r) \geq f(\mathbf{v}_d) \Rightarrow$ Contraction:

$$\mathbf{v}_c = \bar{\mathbf{v}} + \gamma(\mathbf{v}_{d+1} - \bar{\mathbf{v}})$$

with $0 < \gamma \leq 1/2$

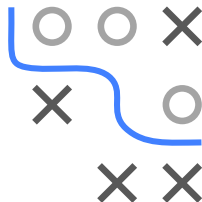
- If $f(\mathbf{v}_c) < f(\mathbf{v}_{d+1})$, accept $\mathbf{v}_c$
- Otherwise, shrink entire simplex (Shrinking):

$$\mathbf{v}_i = \mathbf{v}_1 + \sigma(\mathbf{v}_i - \mathbf{v}_1) \quad \forall i$$

Note: default values for contraction and shrinking coefficients

$$\gamma = \sigma = 1/2$$

4. Repeat all steps until stopping criterion met



NELDER-MEAD SUMMARY

Advantages:

- No gradients needed
- Robust, often works well for non-differentiable functions

Drawbacks:

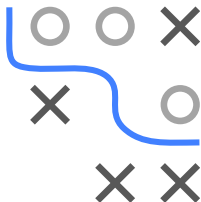
- Relatively slow (not applicable in high dimensions)
- Not each step improves, only mean of corner values is reduced
- No guarantee for convergence to local optimum / stationary point

Visualization:

<http://www.benfrederickson.com/numerical-optimization/>

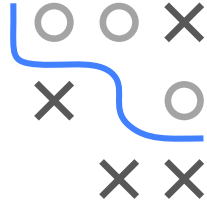
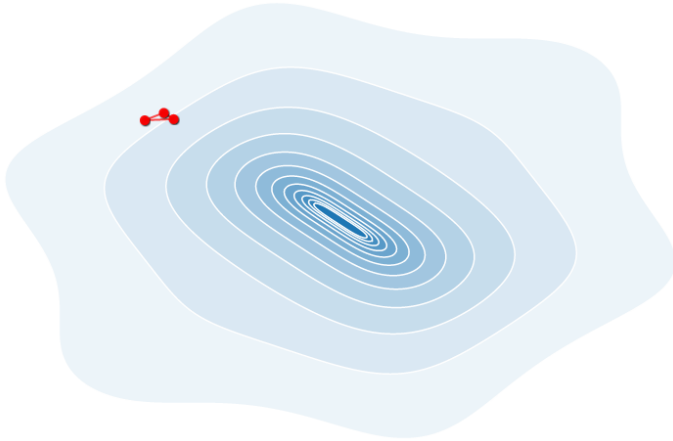
Note: Nelder-Mead is default method of R function `optim()`

If gradient is available and cheap, L-BFGS is preferred



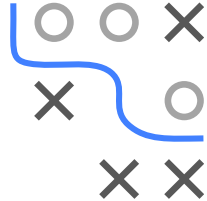
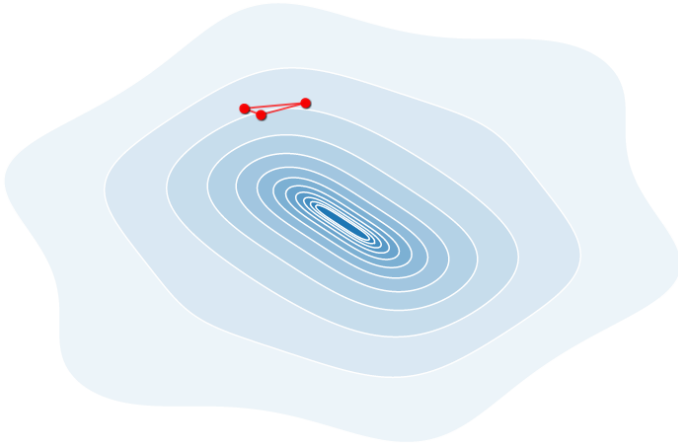
NELDER-MEAD VISUALIZATION IN 2D

$$\min_{\mathbf{x}} f(x_1, x_2) = x_1^2 + x_2^2 + x_1 \cdot \sin x_2 + x_2 \cdot \sin x_1$$



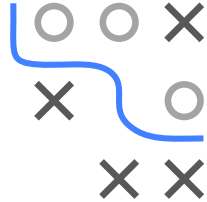
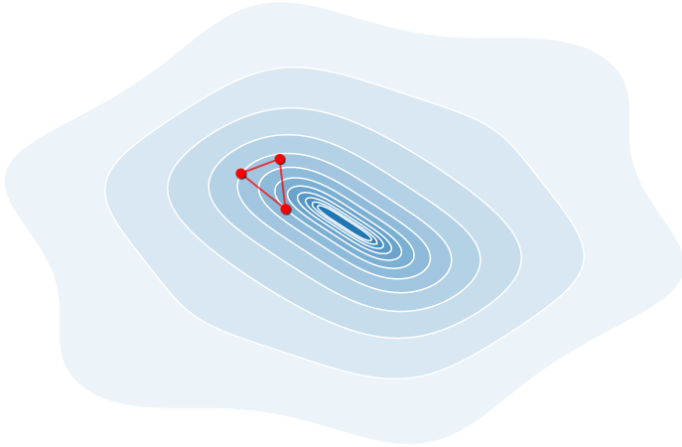
NELDER-MEAD VISUALIZATION IN 2D

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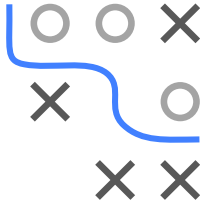
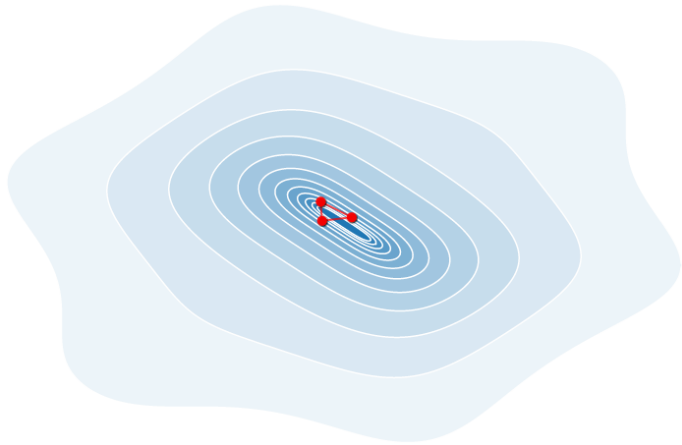
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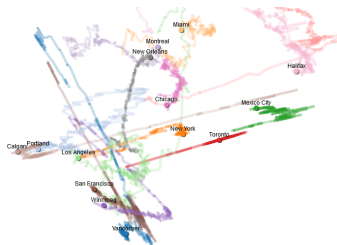
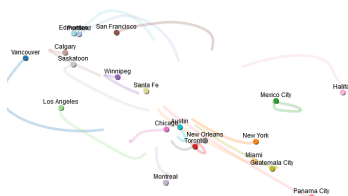
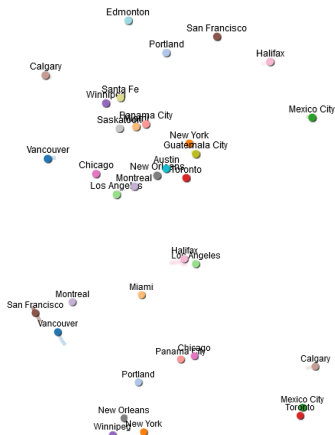
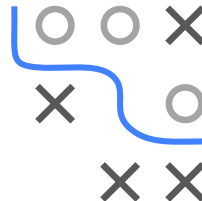


NELDER-MEAD VISUALIZATION IN 2D

$$\min_{\mathbf{x}} f(x_1, x_2) = x_1^2 + x_2^2 + x_1 \cdot \sin x_2 + x_2 \cdot \sin x_1$$



NELDER-MEAD VS. GD



Nelder-Mead in multiple dimensions: Organize points (US cities) to keep predefined mutual distances. For 10 cities, gradient descent (top) converges well for a suitable learning rate. Nelder-Mead (bottom) fails to converge, even after many iterations.

The figure consists of three network diagrams illustrating flight paths between five cities: Vancouver, Portland, New York, Miami, and Mexico City. The cities are represented as nodes, and the flight paths are represented as colored lines (blue, orange, green).

- Top Diagram:** Shows a simple path from Vancouver to Portland to New York. A blue line connects Vancouver and Portland, and an orange line connects Portland and New York.
- Middle Diagram:** Shows a more complex path from Vancouver to Portland to Mexico City to New York. A blue line connects Vancouver and Portland, a green line connects Portland and Mexico City, and an orange line connects Mexico City and New York.
- Bottom Diagram:** Shows a highly complex, multi-branching network connecting all five cities. The network is composed of many overlapping lines of various colors (blue, orange, green) connecting the nodes in a dense, interconnected manner.

A 3x3 grid with a blue path starting at the top-left cell (0,0) and ending at the bottom-right cell (2,2). The path consists of the following cells: (0,0), (0,1), (1,1), (1,2), and (2,2). The cells (0,2), (1,0), and (2,0) are empty. The cells (0,1), (1,1), and (1,2) contain a grey 'X'. The cells (1,0) and (2,0) contain a grey circle.